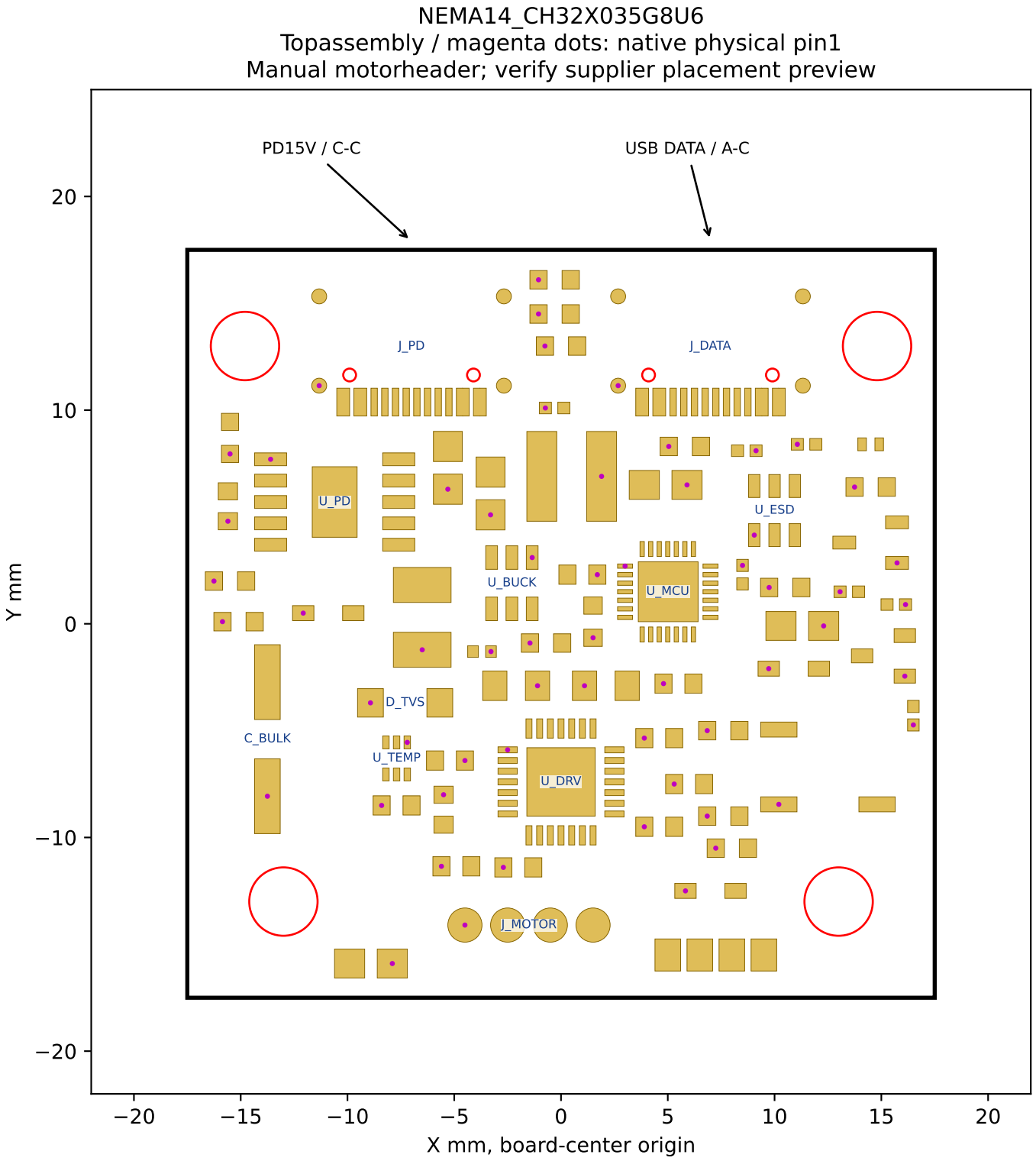


Motor: 1 BlackA+ / 2 GreenA- / 3 RedB+ / 4 BlueB-
PCB NPTH lower26mm / upper29.6mm pitch; front motor26mm pattern is separate.
Cap+ and every diode/IC orientation must match native pads/netlist and supplierpreview.
Filled/capped via list: via_process.csv. See fabrication-notes and mechanical-fit.
SHA256 645d9b85de8dc88191b946ef59a1315769fbab5ee8ad2b12a2c3d1d56374ec45

NEMA14_CH32X035G8U6 - fitted placement reference
Native source convention; supplier pickup/rotation approval pending

Ref	Exact MPN	Native center mm	Rotation	JLCPCB
J_PD	TYPE-C-31-M-12	-7.000,12.975	180	C165948
J_DATA	TYPE-C-31-M-12	7.000,12.975	180	C165948
U_PD	CH224K	-10.600,5.700	270	C970725
R_PD	ERJPA3F1001V	-15.500,2.000	0	C441922
C_PD	CL10A105KB8NNNC	-15.600,5.500	90	C15849
R_PG	0603WAF1002T5E	1.500,0.100	90	C25804
U_MCU	CH32X035G8U6	4.999,1.500	270	C7437027
C_MCU	CC0603KRX7R9BB104	1.000,2.300	180	C14663
C_LOGIC	CL21A475KBQNNNE	-8.900,-15.900	180	C98192
R_CC1	0603WAF5101T5E	5.800,8.300	0	C23186
R_CC2	0603WAF5101T5E	0.000,13.000	0	C23186
R_USB_SENSE_H	0603WAF1002T5E	14.500,6.400	0	C25804
R_USB_SENSE_L	0603WAF1002T5E	-0.700,-0.900	0	C25804
Q_DATA	MMBT3904-7-F	14.500,3.800	0	C94514
R_VM_H	0603WAF1003T5E	-0.300,14.500	0	C25803
R_VM_L	0603WAF1002T5E	-0.300,16.100	0	C25804
R_VM_BLEED	0603WAF4700T5E	-15.100,0.100	0	C23179
C_VM_SENSE	CL10B103KB8NNNC	5.500,-2.800	0	C1589
U_BUCK	AP63203WU-7	-2.300,1.900	180	C780769
L_BUCK	SRN4018-4R7M	0.500,6.900	180	C780206
C_BOOTSTRAP	CL05B104KO5NNNC	-3.700,-1.300	180	C1525
C_BUCK_IN	CL32B106KBJNNNE	-6.500,0.300	90	C138687
C_BUCK_OUT1	CL21A226MAQNNNE	-5.300,7.300	90	C45783
C_BUCK_OUT2	CL21A226MAQNNNE	-3.300,6.100	90	C45783
C_BUCK_OUT3	CL21A226MAQNNNE	4.900,6.500	180	C45783
D_PD	1N5819WS	-10.900,0.500	0	C191023
D_DATA	1N5819WS	7.000,-12.500	0	C191023
U_DRV	A4988SETTR-T	0.000,-7.400	270	C38437
C_CP	CC0603KRX7R9BB104	-5.500,-8.700	270	C14663
C_VCP	CC0603KRX7R9BB104	-4.900,-11.350	0	C14663
C_VREG	CL10B224KA8NNNC	-2.000,-11.400	0	C21120
C_DRV_LOGIC	CC0603KRX7R9BB104	4.600,-9.500	0	C14663
C_VM	CC0603KRX7R9BB104	-5.200,-6.400	180	C14663
C_VM2	CC0603KRX7R9BB104	4.600,-5.350	0	C14663

NEMA14_CH32X035G8U6 - fitted placement reference
Native source convention; supplier pickup/rotation approval pending

Ref	Exact MPN	Native center mm	Rotation	JLCPCB
C_BULK	EEEFPV101XAP	-13.750,-5.400	90	C178585
R_SA	FRL0805FR240TS	2.100,-2.900	0	C2930216
R_SB	FRL0805FR240TS	-2.100,-2.900	180	C2930216
R_REF_H	0603WAF6201T5E	7.600,-9.000	0	C4260
R_REF_L	ERJPA3F1001V	7.600,-5.000	0	C441922
C_REF	CL10B103KB8NNNC	6.000,-7.500	0	C1589
R_ENABLE	0603WAF1002T5E	8.000,-10.500	0	C25804
R_SLEEP	0603WAF1002T5E	10.500,1.700	0	C25804
J_MOTOR	B4B-PH-K-S(LF)(SN)	-1.500,-14.100	0	C131334
U_ESD	USBLC6-2SC6	10.000,5.300	0	C7519
D_TVS	SMF4L16A	-7.300,-3.700	0	C1973135
R_USB_BOOT	0402WGF4701TCE	11.500,8.400	0	C25900
D_STATUS	19-217/GHC-YR1S2/3T	-15.500,8.700	90	C72043
R_LED	0402WGF1001TCE	-0.300,10.100	0	C11702
U_TEMP	TMP102AIDRLR	-7.700,-6.300	90	C99269
C_TEMP	CC0603KRX7R9BB104	-7.700,-8.500	0	C14663
R_TEMP_SCL	0402WGF5101TCE	8.700,8.100	180	C25905
R_TEMP_SDA	0402WGF5101TCE	8.500,2.300	270	C25905
R_TEMP_ALERT	0402WGF1002TCE	13.500,1.500	0	C25744
BZ_ALARM	MLT-5020	12.500,-6.700	0	C94598
Q_BUZZER	AO3400A	15.100,-1.500	0	C20917
D_BUZZER	1N5819WS	10.900,-2.100	0	C191023
R_BUZZER_GATE	0402WGF1000TCE	15.700,0.900	180	C25076
R_BUZZER_PD	0402WGF1003TCE	16.500,-4.300	90	C25741
C_BUZZER	CL21A226MAQNNNE	11.300,-0.100	180	C45783